



Nanocomposite Organic Scintillators for PET Imaging

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Outline



- › Context
- › How scintillators work
- › What makes a good scintillator
- › Scintillator landscape
- › Motivation for nanocomposites
- › Progress and Challenges



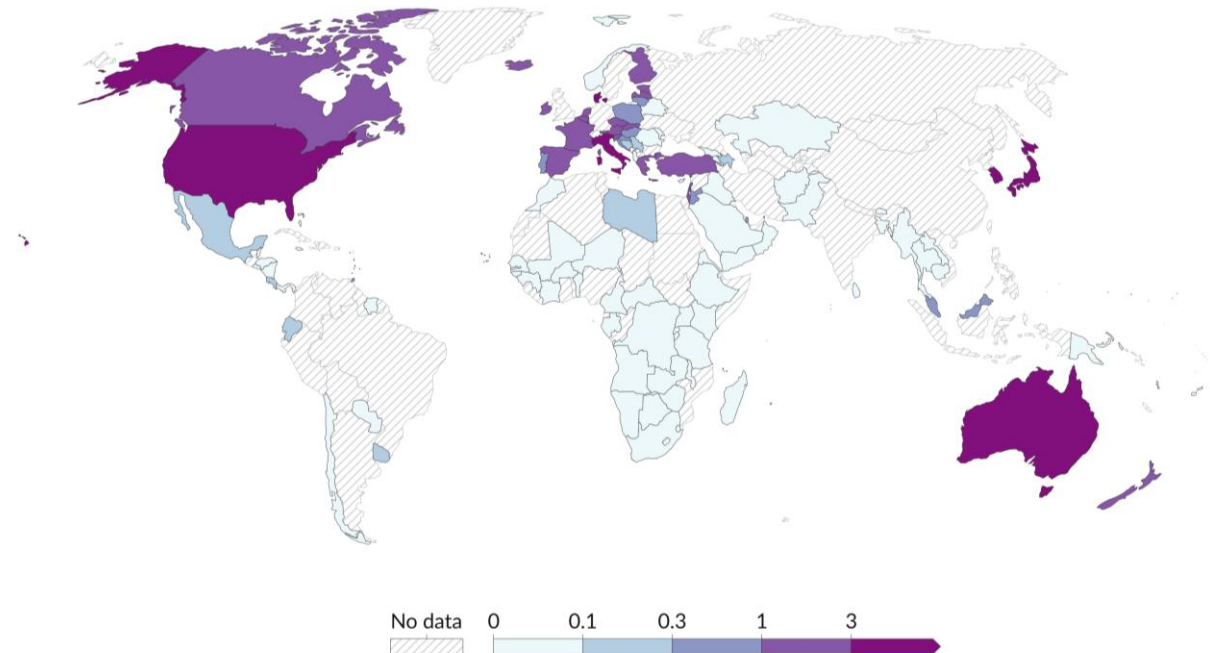
The Big Question

- › How do we make PET cheaper and more accessible?
- › Make cheaper scintillators?

Positron emission tomography (PET) imaging units per million people, 2021

Number of PET¹ units, devices for advanced imaging using radioactive tracers, per million people in the population.

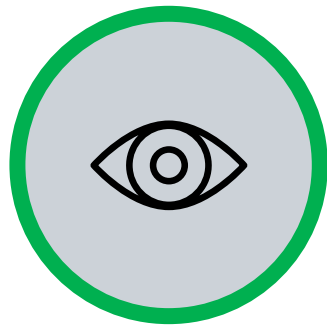
Our World
in Data



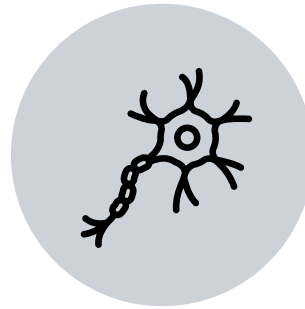
Data source: World Health Organisation (2022)

OurWorldinData.org/cardiovascular-diseases | CC BY

What scintillators do



Detection

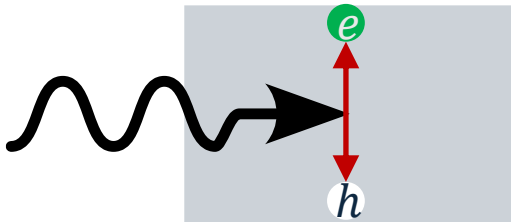


Transmission

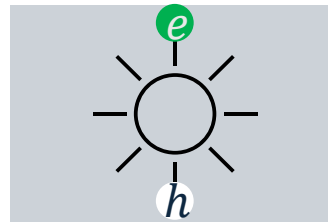


Processing

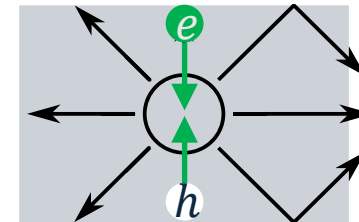
How scintillators work



Absorption

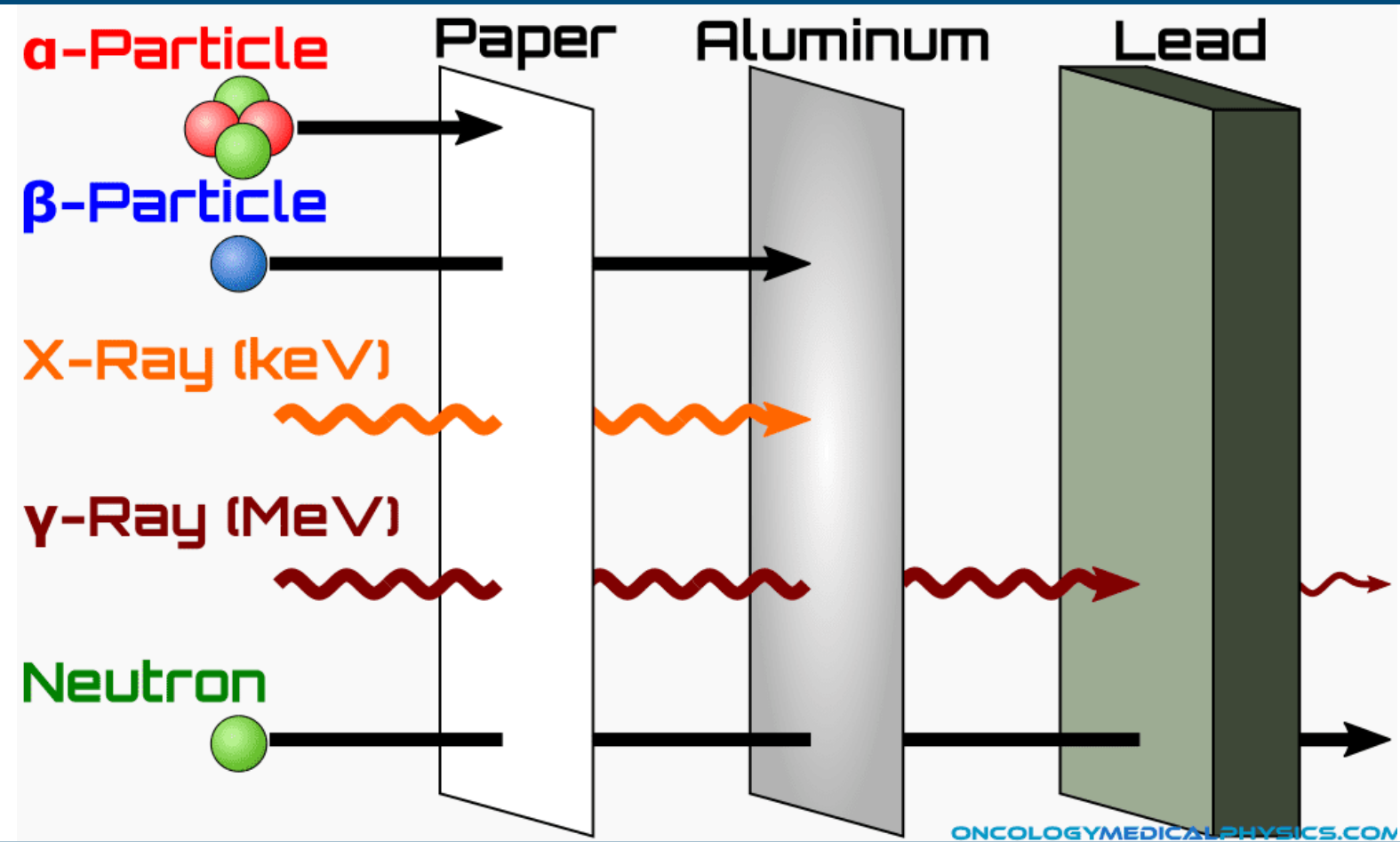


Energy Transfer



Luminescence

Stopping γ photons is not trivial



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Absorption

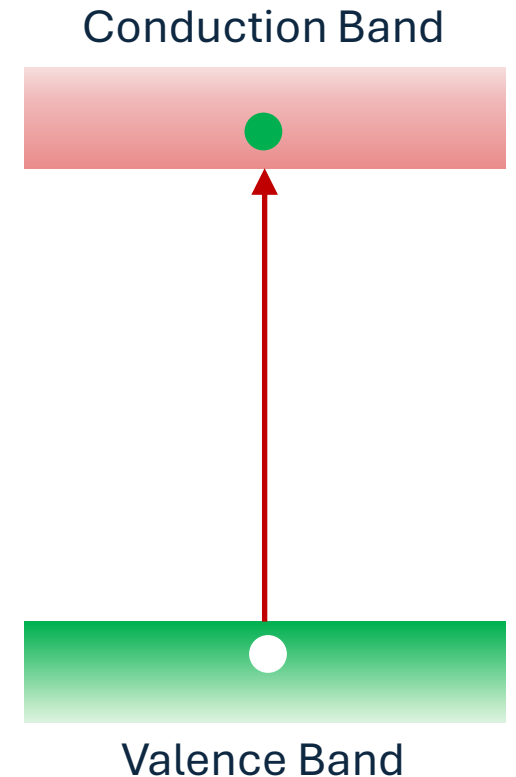
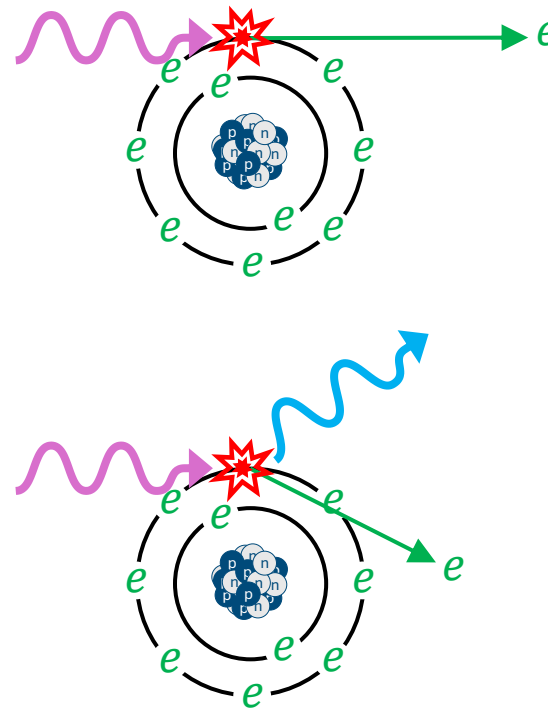
- Photoelectric Effect

$$\sigma_p \approx \frac{Z^3}{E^3}$$

- Compton Scattering

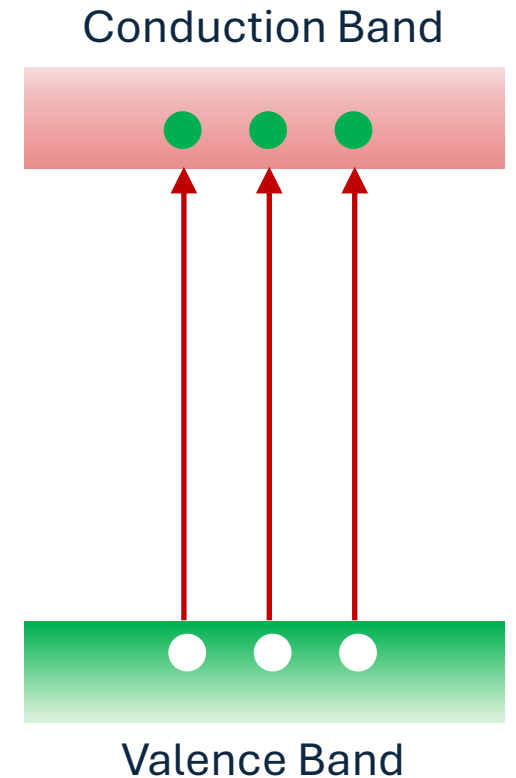
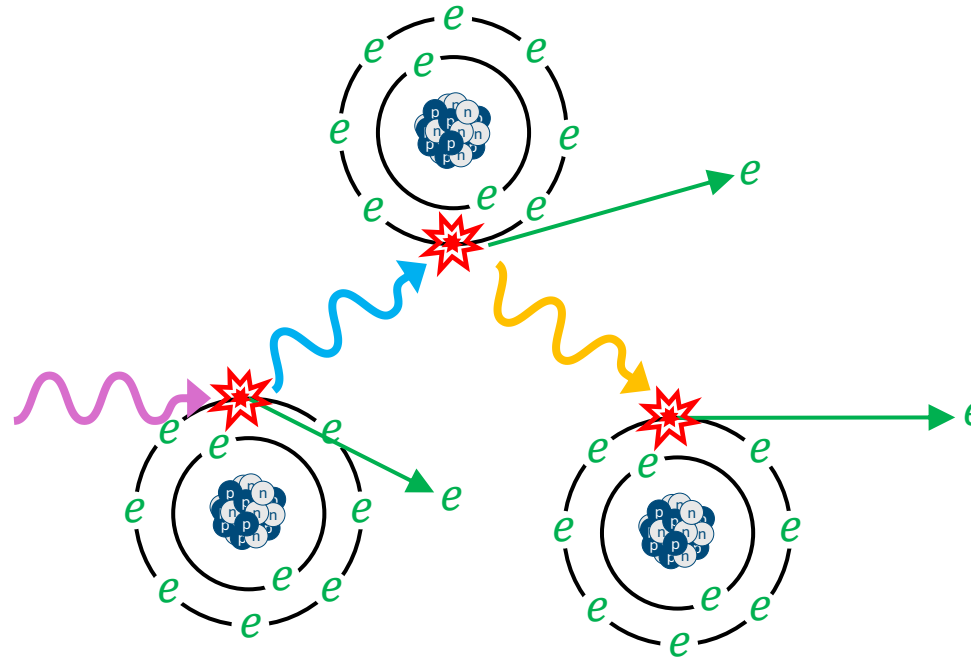
$$\sigma_c \approx \rho_e$$

- Exciton creation



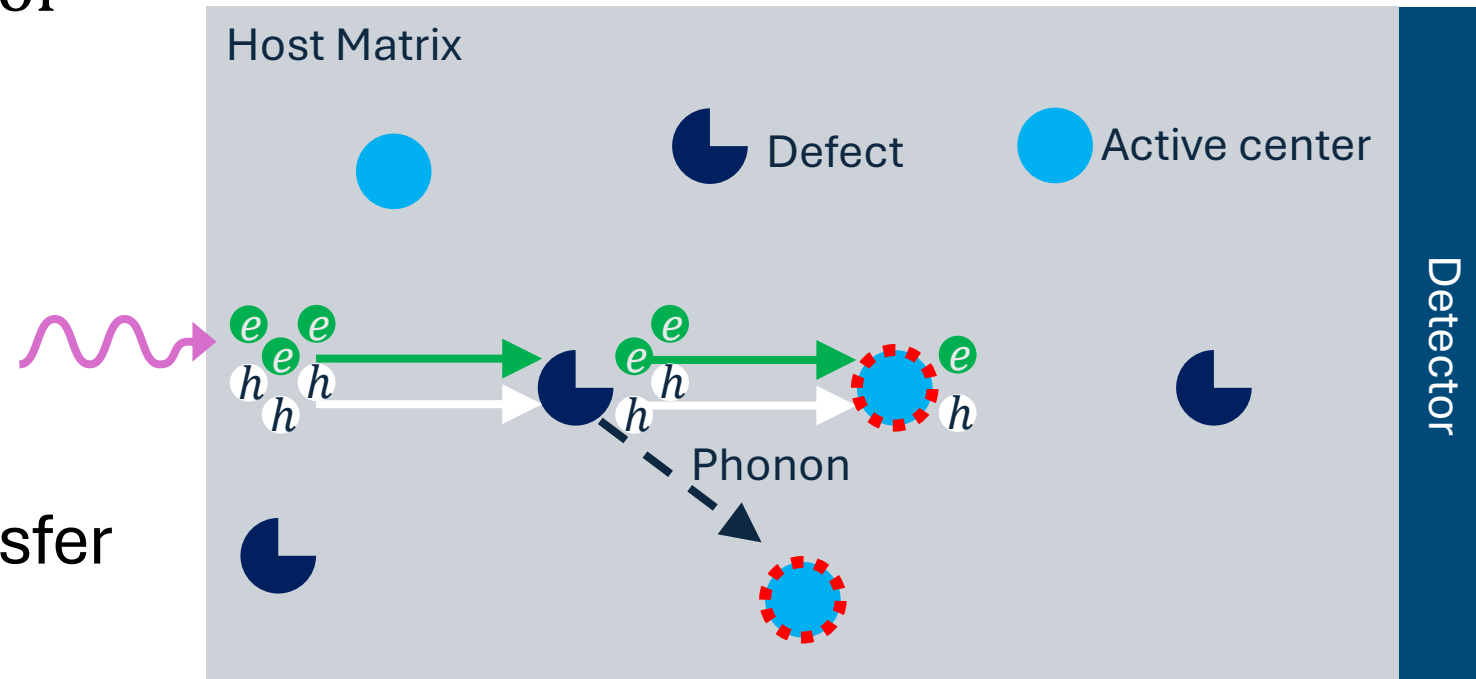
Absorption

- › Secondary scatters
- › γ energy depletion
- › Exciton cascade



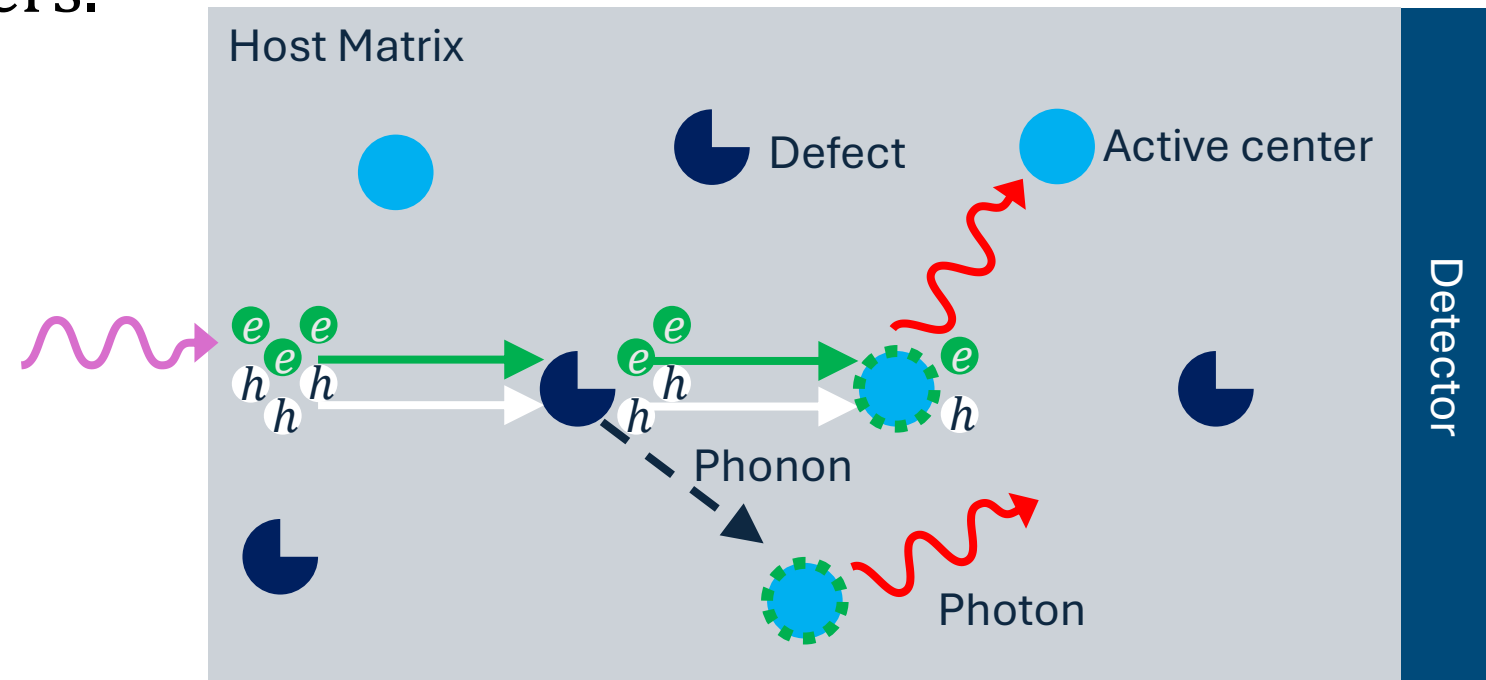
Energy Transfer

- › Non-radiative *excitation* of active centers
 - › Exciton recombination
 - › Lattice vibrations (heat)
- › Förster/Dexter energy transfer



Luminescence

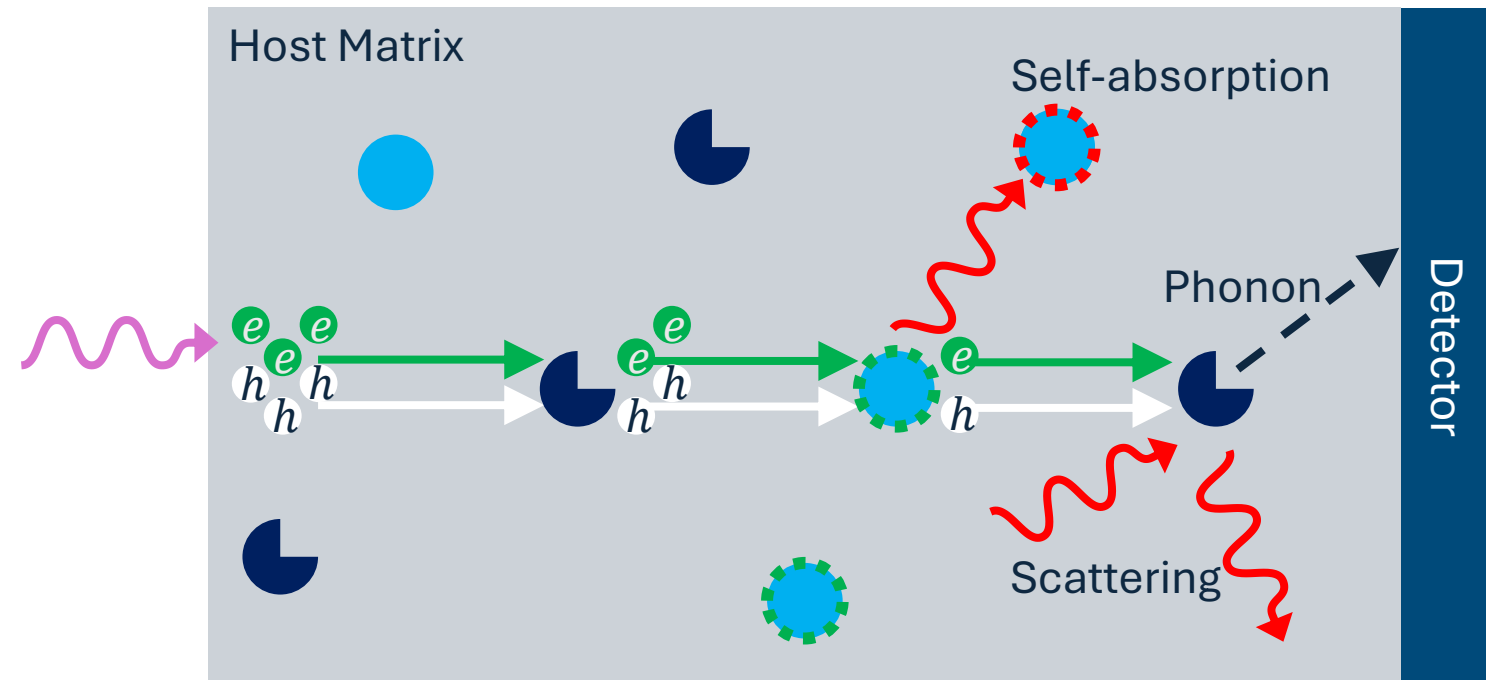
- › *Relaxation* of active centers.
- › *Photon* emission.
- › Transmitted out.



Losses



- › Phonon emission
- › Self-absorption
- › Scattering



What makes a good scintillator



› Light Yield

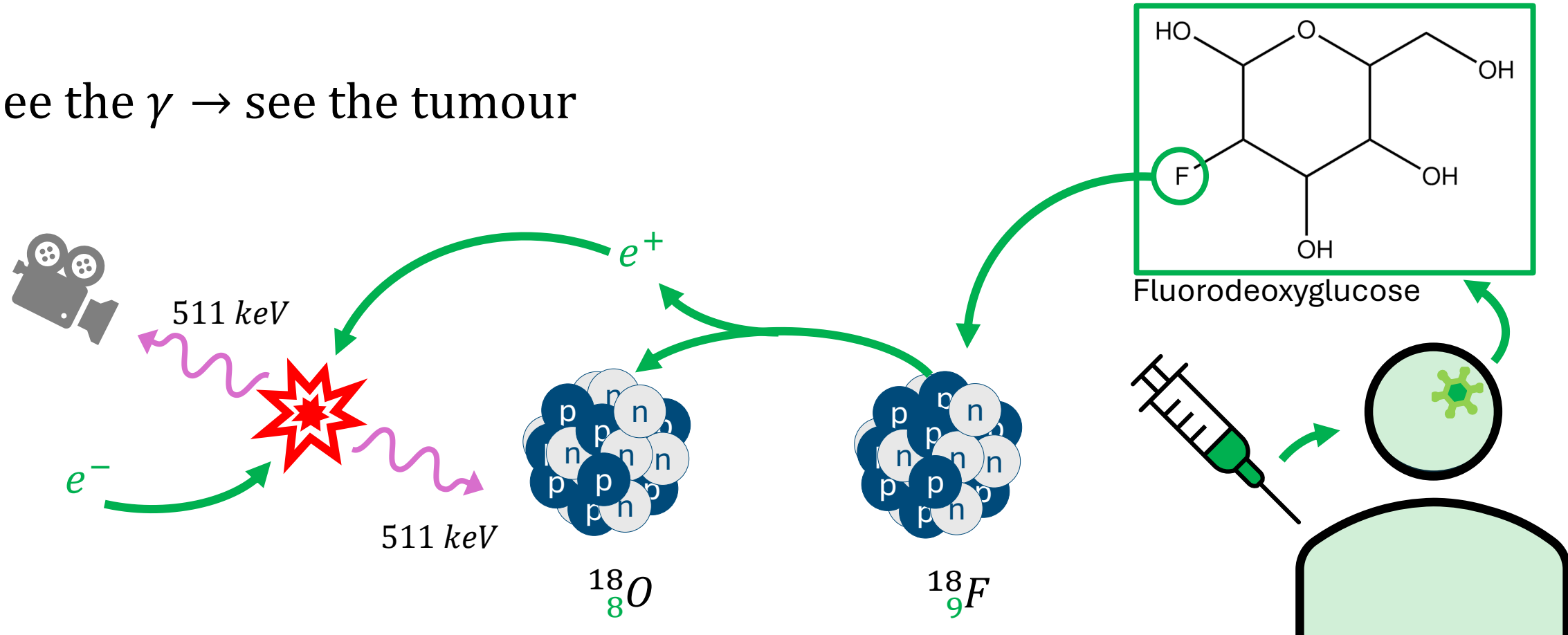
› Fast Scintillation

› Energy Resolution

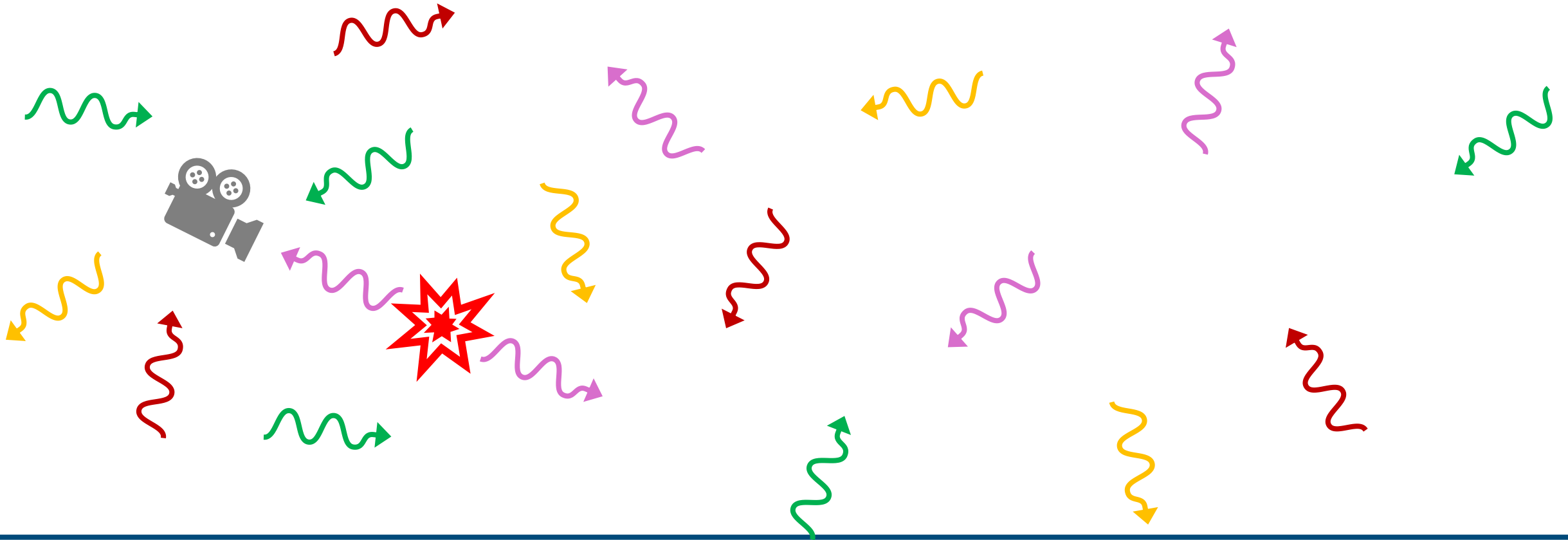


Recap

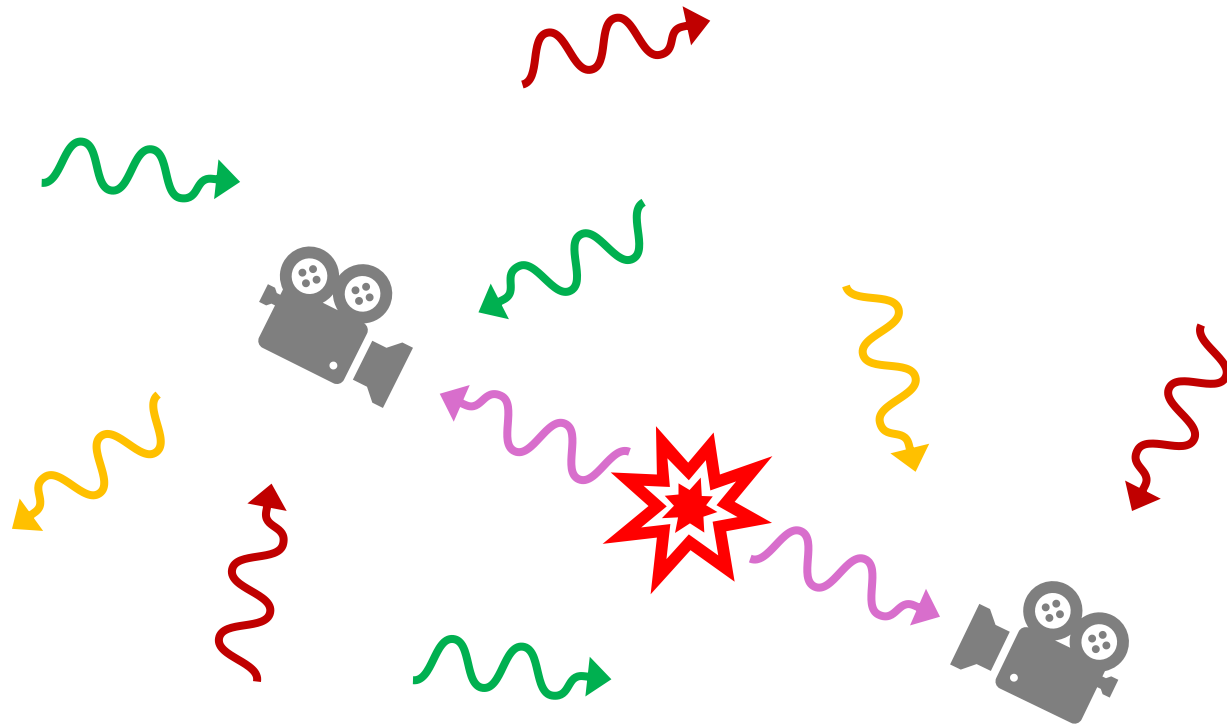
› See the γ → see the tumour



Noisy complications



Noise Elimination

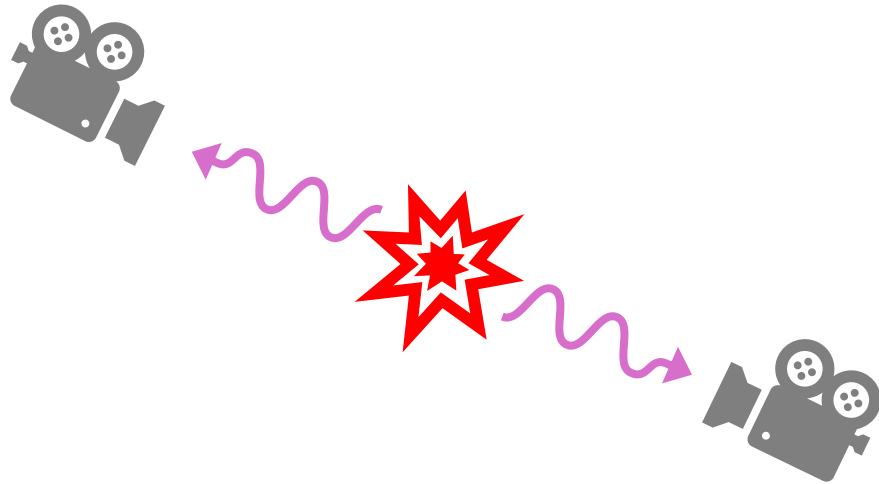


- › Add another camera
- › Coincidence timing
- › Fast scintillation

Noise Elimination

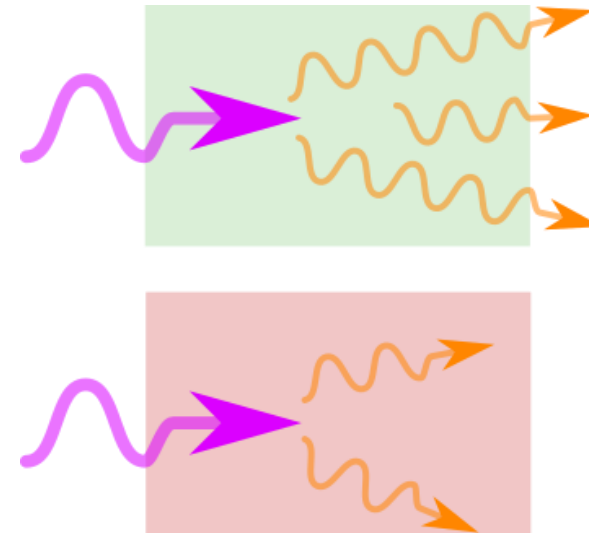


- › Add another camera
 - › Coincidence timing
 - › Fast scintillation
- › Filter by Energy
 - › Photopeak identification
 - › Energy resolution



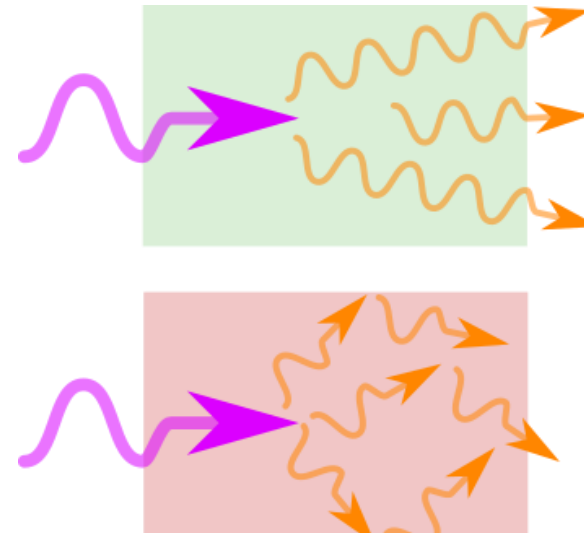
Light Yield

- › High stopping power
 - › Density
 - › Atomic number
- › High transparency
 - › Low self-absorption
 - › Minimal scattering



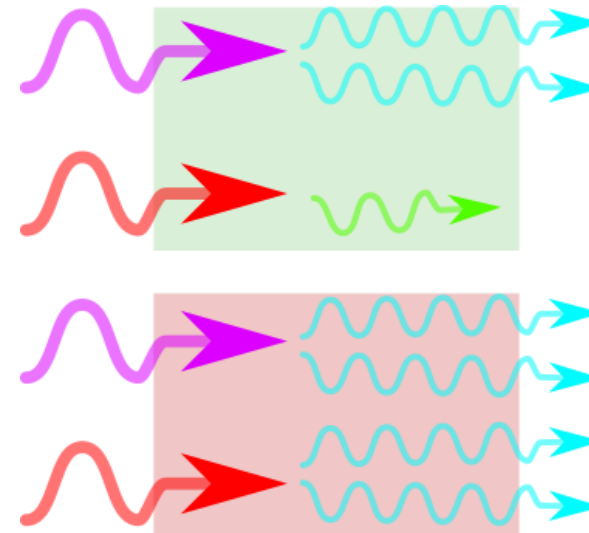
Fast scintillation

- › Fast exciton decay
 - › Short recombination length
 - › Concentrated active centers
- › Efficient FRET/DET
 - › Minimal energy transfer paths
 - › Strong quantum resonance



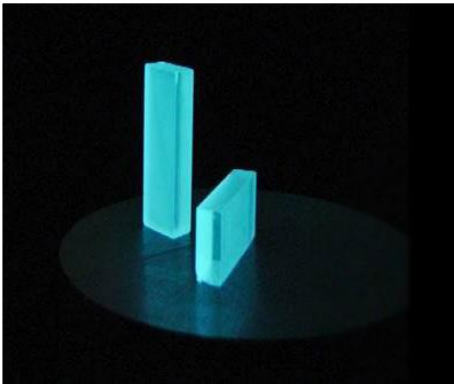
Energy Resolution

- › High stopping power
 - › Reduce radiation escape
- › Photoelectric interaction σ
 - › High Z
 - › $\sigma_{pe} \approx \frac{Z^3}{E^3}$



Scintillator Landscape

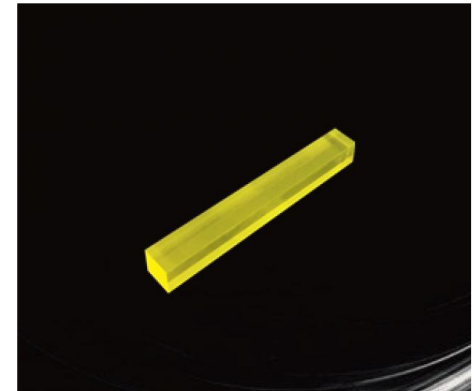
› Crystals



› Organics



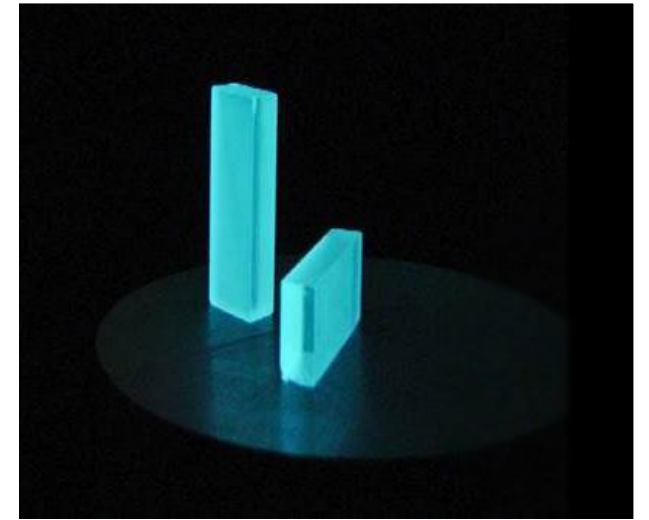
› Ceramics



Crystals



- › **Stopping power** – 1 cm attenuation (high Z)
- › **Energy resolution** – photoelectric events
- › **Fast scintillation** – 40 ns decay time
- › **Transparent** – 32 000 photons/MeV light yield
- › **Expensive**



Organics



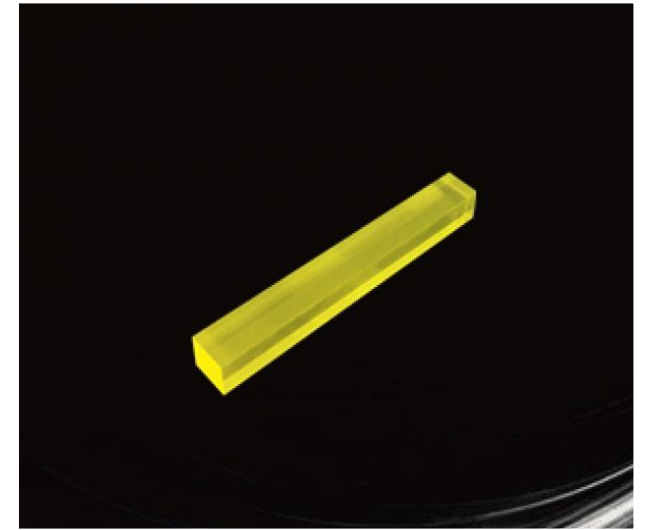
- › **Stopping power** – 10 cm attenuation (low Z)
- › **Energy resolution** – NO photoelectric events
- › **Fast scintillation** – 2 ns decay time
- › **Transparent** – 10 000 photons/MeV light yield
- › **Expensive**



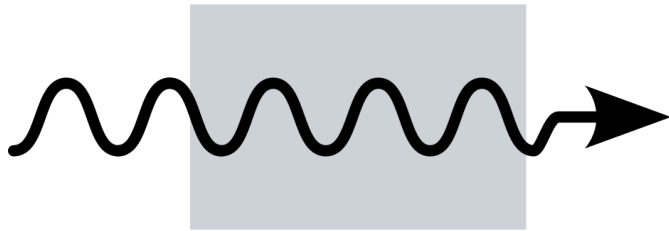
Ceramics



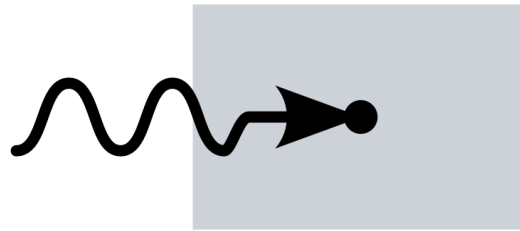
- › **Stopping power** – 10 cm attenuation (high Z)
- › **Energy resolution** – photoelectric events
- › **Slow scintillation** – $1 \mu\text{s}$ decay time
- › **Opaque** – 20 000 photons/MeV light yield
- › **Inexpensive**



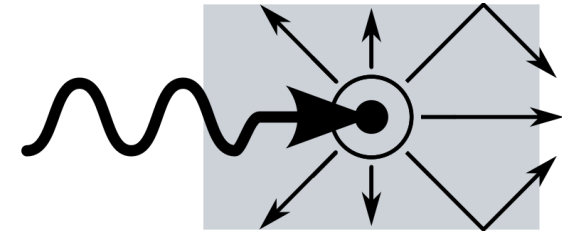
Motivation for Nanocomposites



Cheap polymer
host matrix



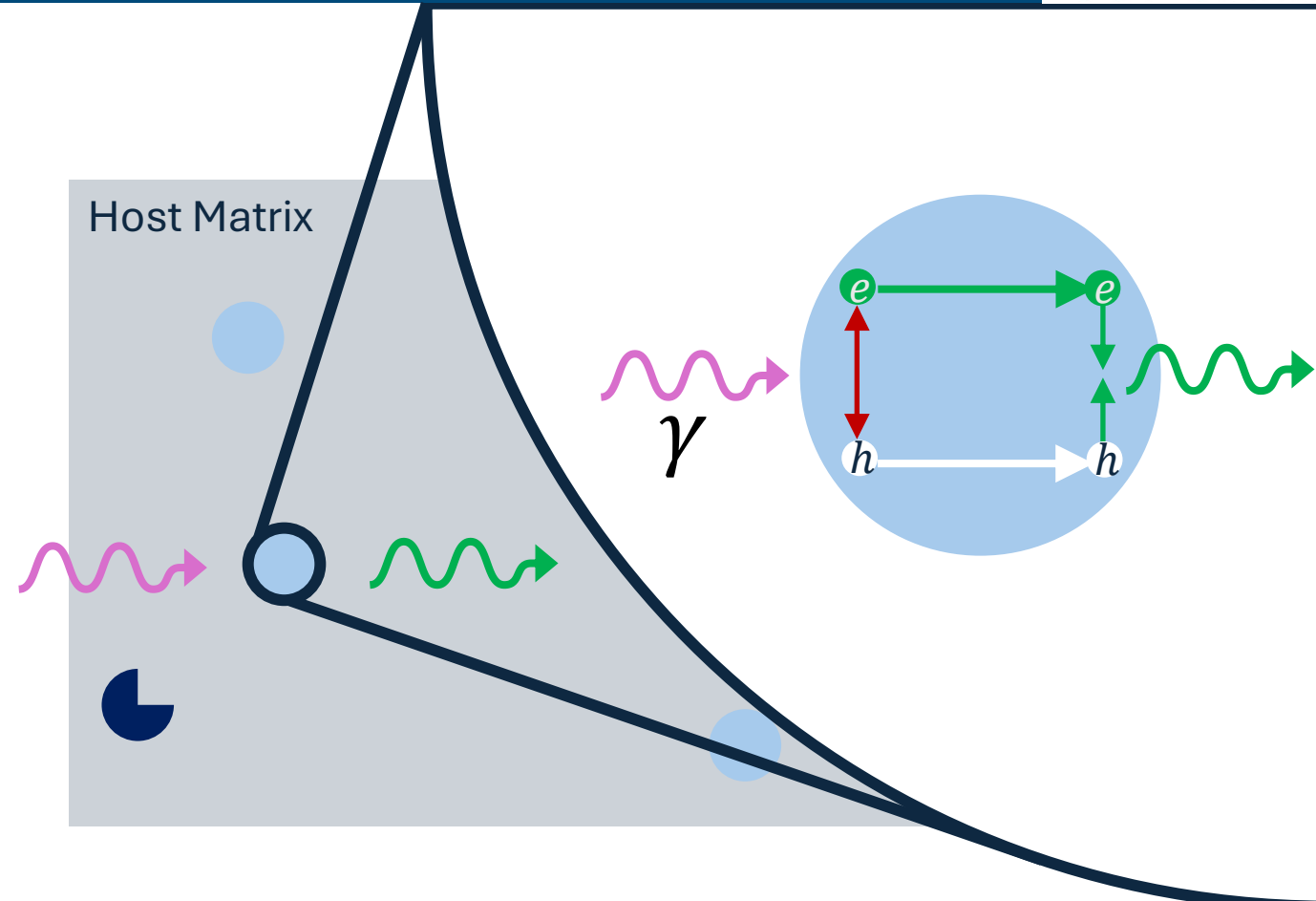
Small, dense
activation centres



Fast scintillation

Nanoparticle Activation Centers

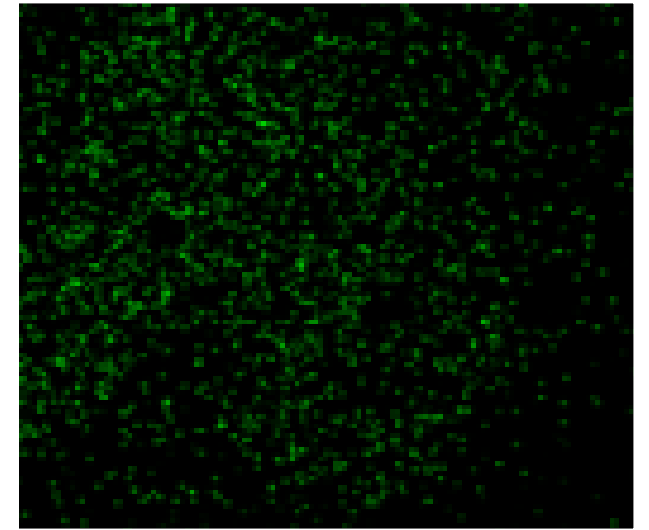
- › Spatially confined excitons
 - › Fast scintillation
- › High Z
 - › Large γ cross-section
 - › Photoelectric interactions
- › Tuneable excitation/emission



Nanocomposite plastic



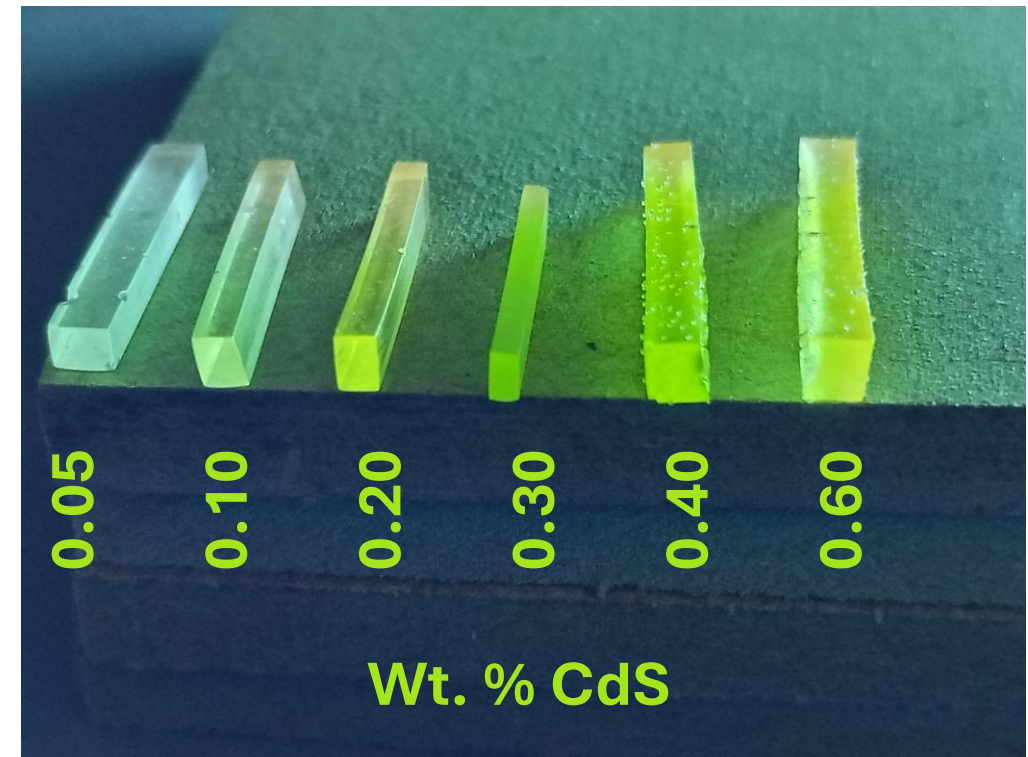
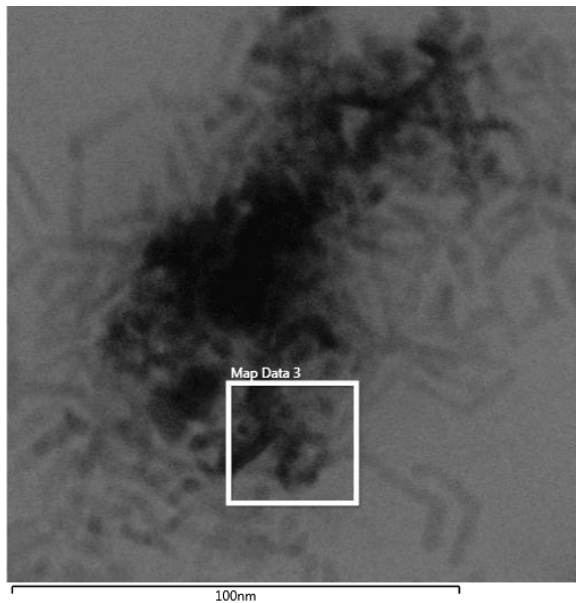
- › Stopping power – High Z
- › Energy resolution – photoelectric events
- › Fast scintillation – 1 ns decay time
- › Transparent? – High self-absorption and scattering
- › Inexpensive?



Trial #1 – Epoxy Resin + CdS Nanoplatelets

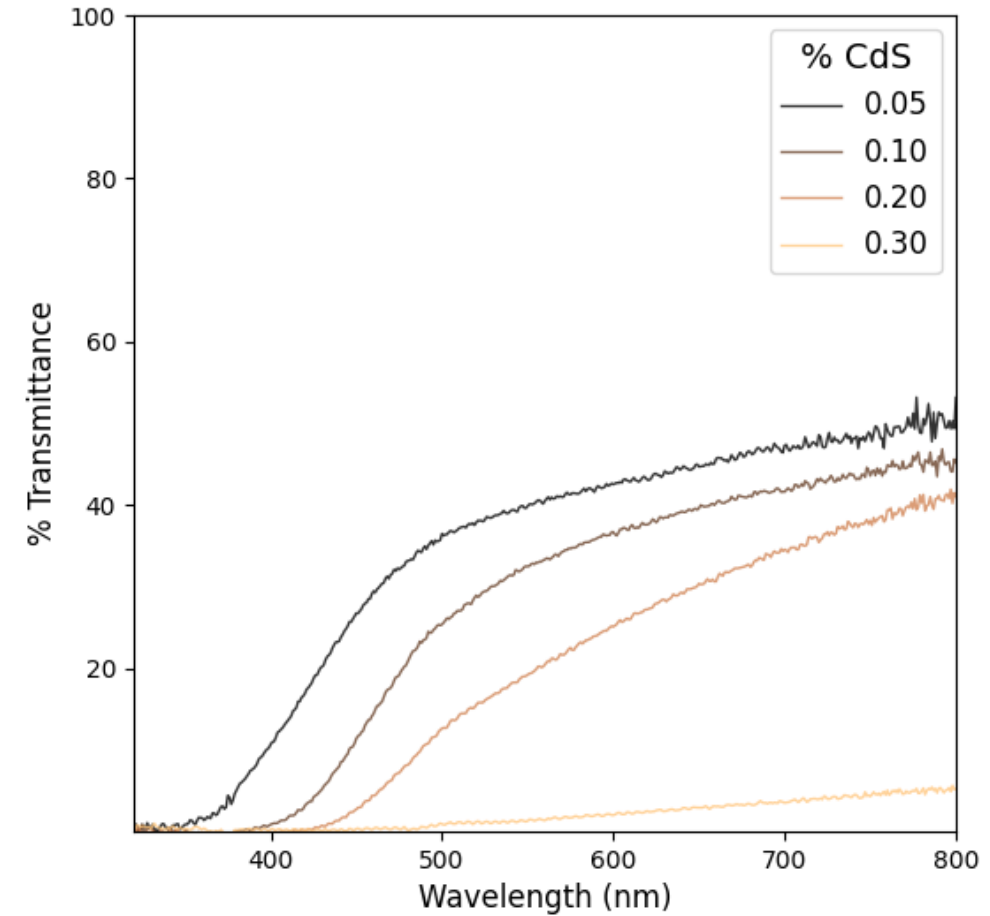
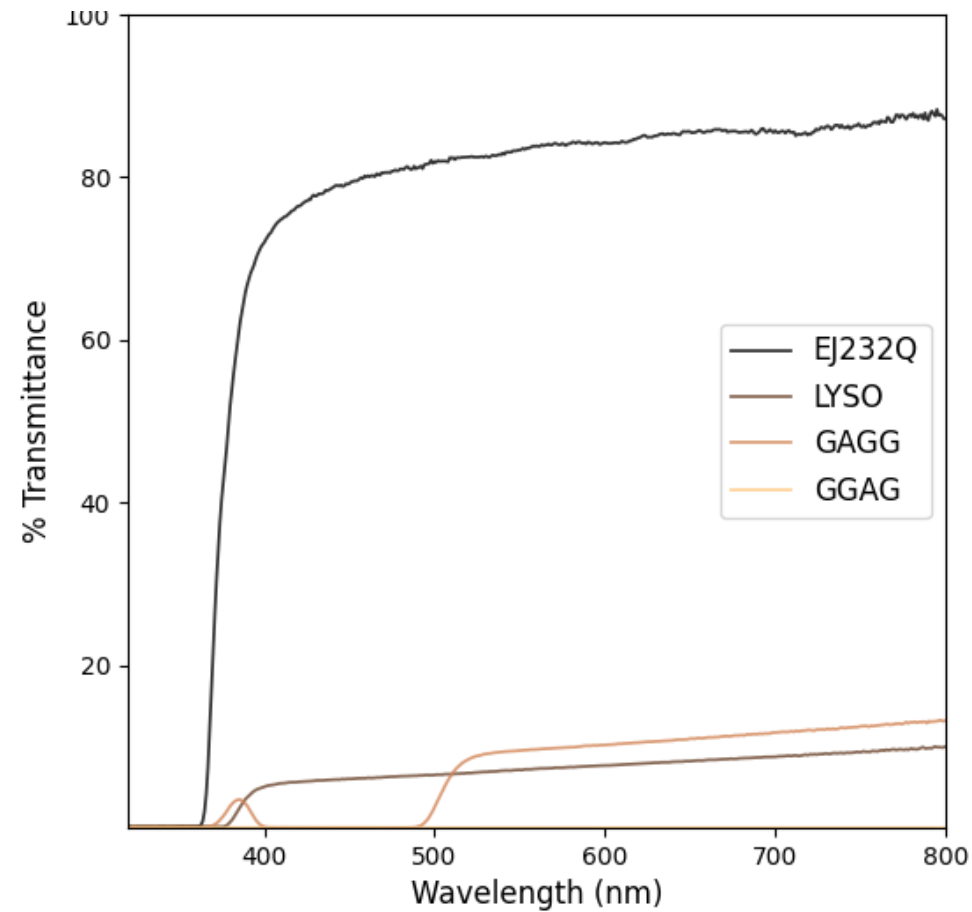


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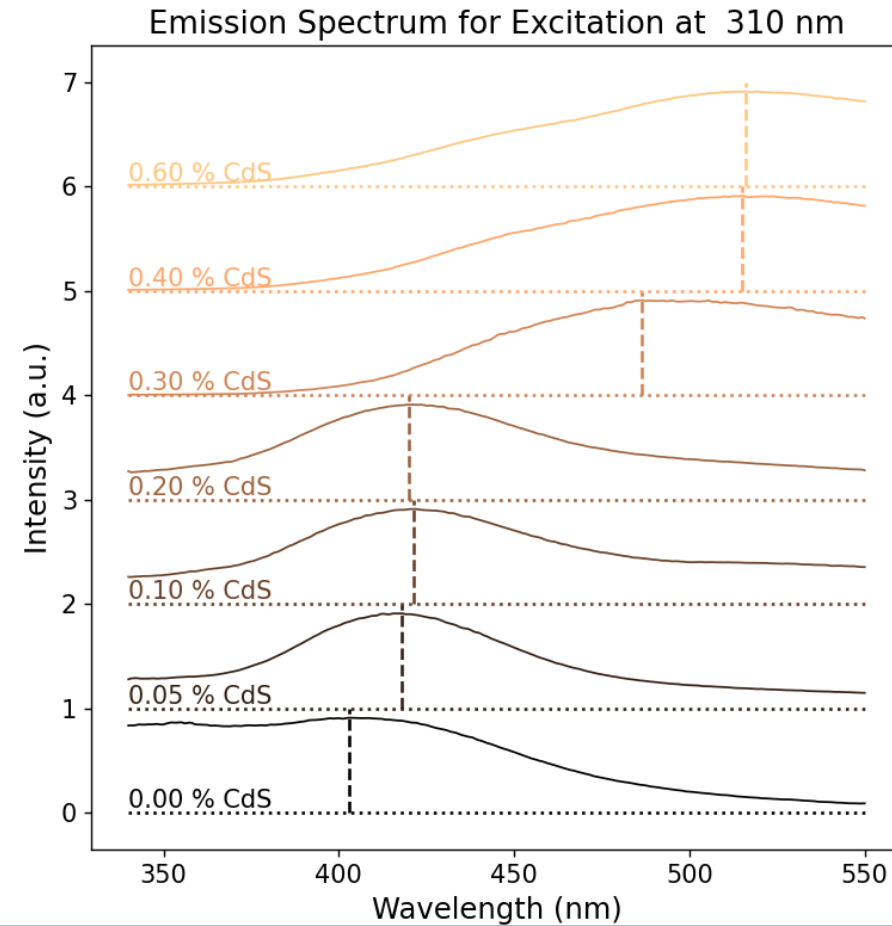


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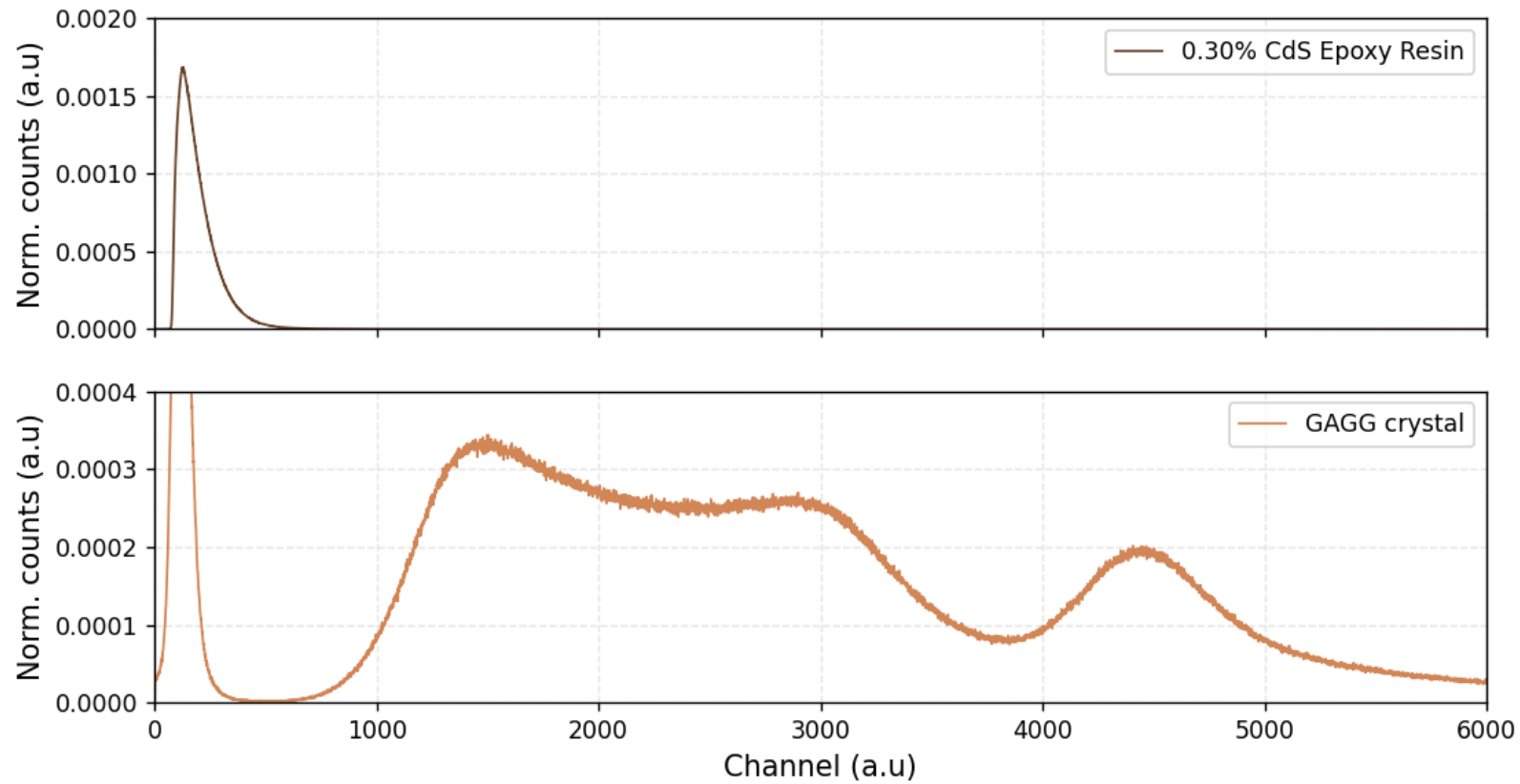
Transmittance



Photoluminescence



Light Yield



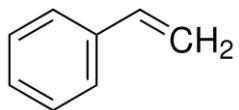
Problems and Potential Solutions

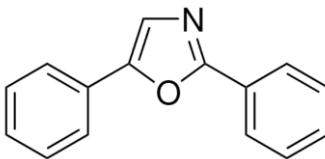


- › Low transmittance
 - › Nanoparticle aggregation
 - › Poor polymer integration
- › Low light yield
 - › Inefficient energy transfer
 - › Inadequate active centers
 - › High self absorption
- › Surface Ligand Exchange
 - › Co-polymerizable coupling agents
 - › Mitigate defect formation
- › Conventional dopants
 - › Secondary active centers – PPO
 - › Wavelength shifter - POPOP

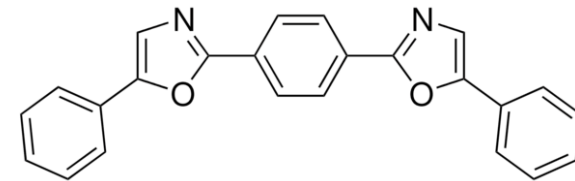


Trial #2 – Conventional Polystyrene

› Styrene 

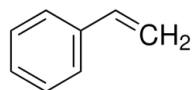
› 2,5-Diphenyloxazole (PPO) 

› 1,4-Bis(5-phenyl-2-oxazolyl)benzene (POPPOP)

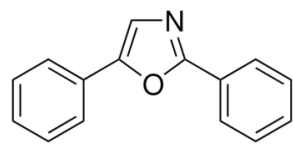


Trial #2 – Conventional Polystyrene

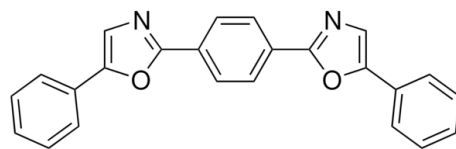
› Styrene



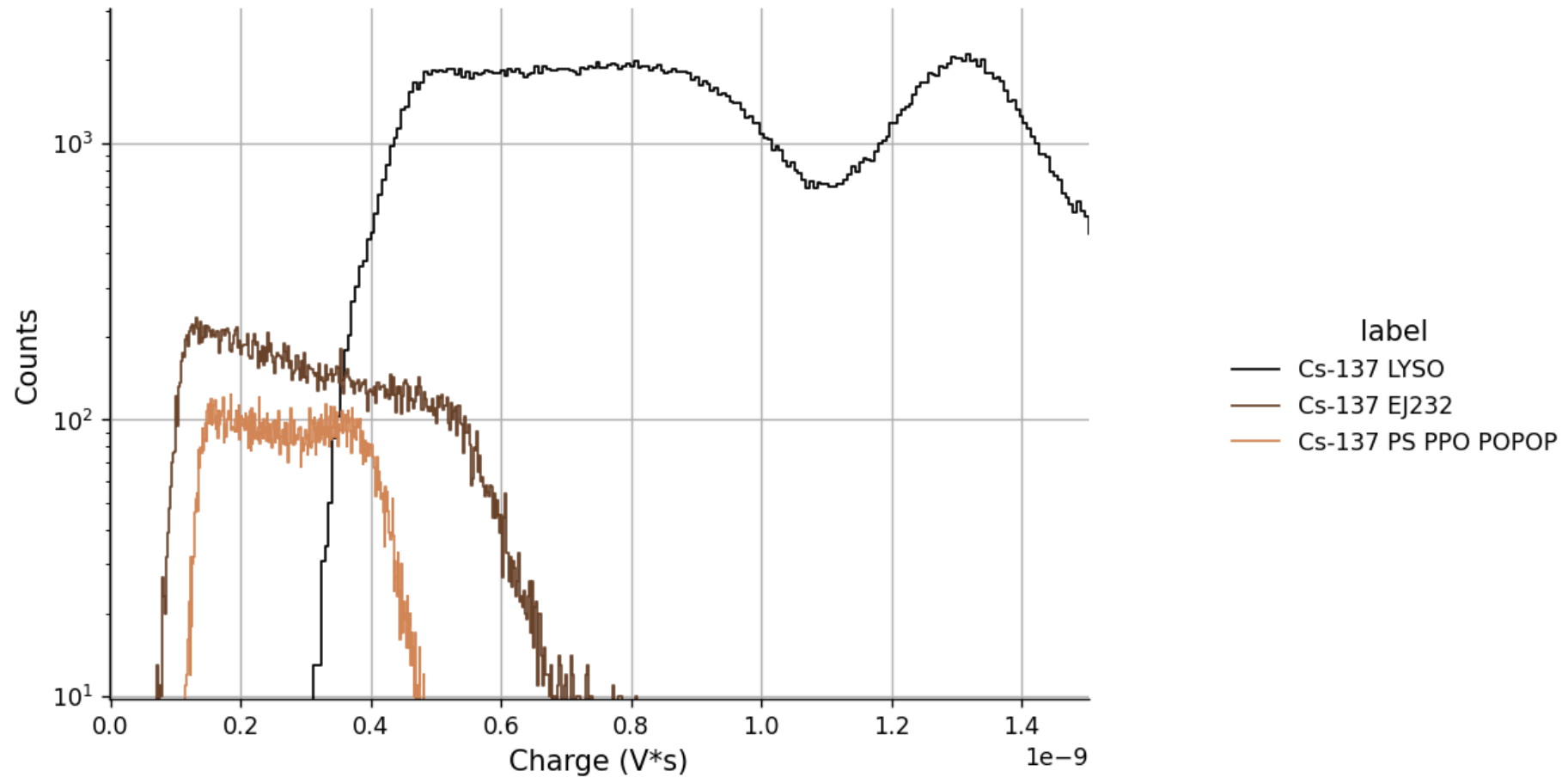
› PPO



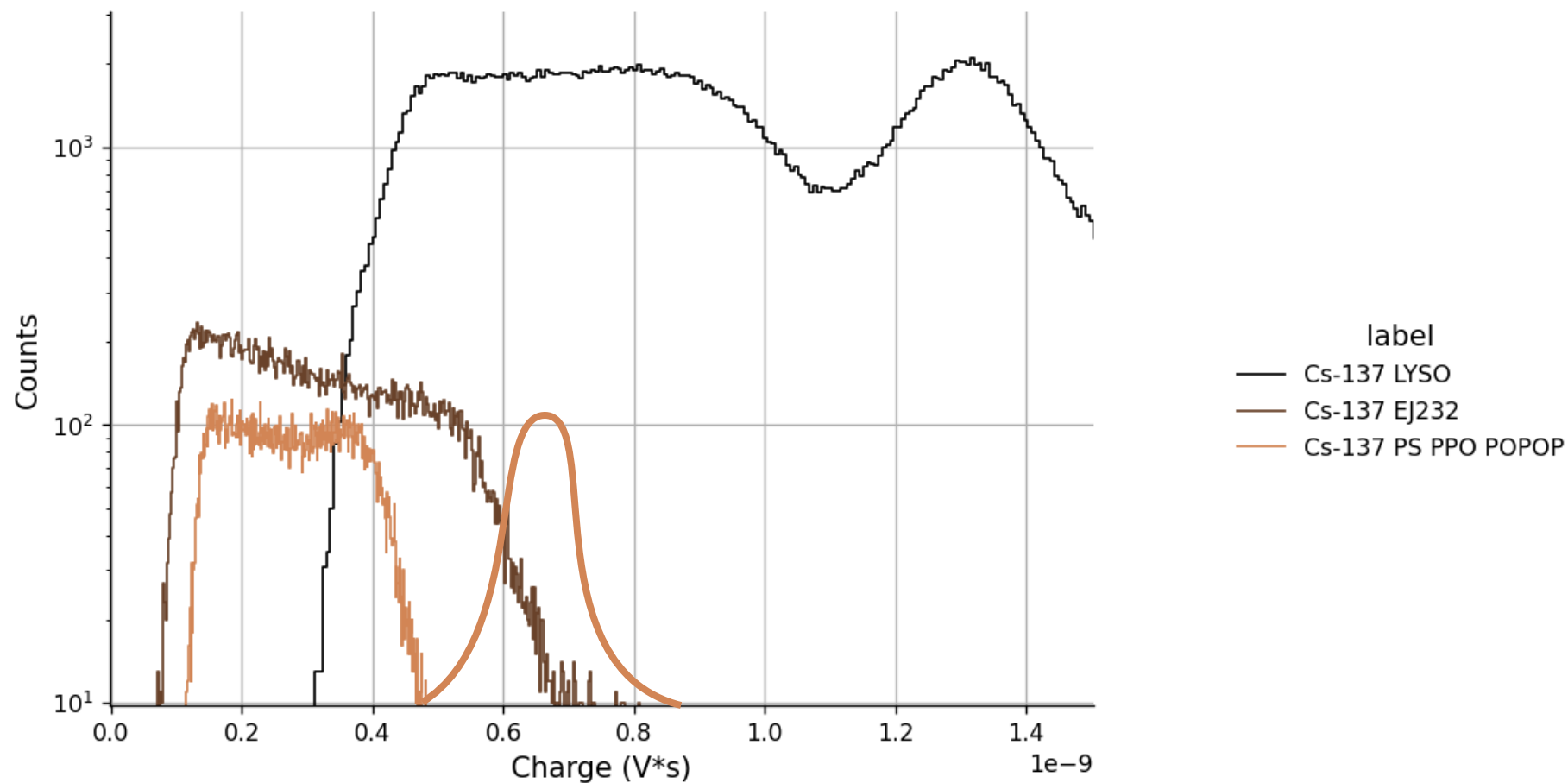
› POPOP



Energy spectrum



Trial #3 – Add CdSe nanoparticles





Thank you!